

Predicting customer buying behaviour

summary of the key findings from the customer booking analysis:

- **Data Overview:** The dataset was clean with no missing values, but required some preparation due to categorical data.
- **Predictive Model:** We trained a model to predict customer bookings, achieving reasonable overall accuracy (around 85%).
- **Model Performance:** While accurate overall, the model needs improvement in identifying actual bookings due to the low number of bookings in the data.
- **Key Factors:** The most important factors influencing bookings are how far in advance a customer books, the time of day of the flight, the length of the stay, and the flight duration.
- As shown in the chart, '**purchase_lead**', '**flight_hour**', '**length_of_stay**', and '**flight_duration**' are the most influential features in predicting customer bookings

Feature Importance Visualization:

